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Initial Meeting 9/8

Project Summary

- The program has been around for 20+ years. Designed to teach spherical geometry. Thinking about shortest distant paths, by analyzing what is happening in the plane. Taking a straight line onto a circle. Feedback has been received about being overwhelmed with all of the buttons. The point of the project is to create tours to guide users through one specific action. Mostly for use in geometry classes. Parametric curves or calculation features may be in the cards if time allows.
- github.com/dulimarta/spherical-easel
- Tooltip is used for the UITour.vue file. It is possible for a single tooltip to be attached to many elements on the UI. The tooltip is to
- The ID is given to each unique component of the different tabs. The name of the tab is the name of the class in the HTML code. The ID is needed for the anchor to work.
- The four files that are in the the most recent commit on github, will be our workspace.

Goal:

- Dulimarta: **Adding a tour layer separate from the main application. Potentially having a record mode** that can be used to create a tutorial/tour to be shared with their students. Could be saved in Firebase. It is tedious to write out all of the scripts
- Instead of having a 'next' button, it would be nice to be able to have some kind of interaction that can take input based on what it asks to do. A button for the first version of implementation, we could use a next button.

End of Semester Goal:

- Would love to be able to move through a couple of tours.
- Building the mechanism to create tours would be the best way. Documentation is a MUST.
- A recordable tour would be amazing, but might not be feasible for the semester.
- The app should be internationalized, but it is not the main focus of what we are to do. It would be nice if we kept it in mind.

Process, Methodology, Roles and Logistics

- 9AM-10AM Tuesdays every week @ C-2204
- We will use keys from a map that will show messages, to be able to translate to other languages in the web app.
 - Uses the T Function (translate)
- Users range from Middle Schoolers all the way to scholarly researchers.

UI

Which features should the first tours focus on?

What would be the highest Priority Tours first?

- The first tutorial/tour we should do is “how to to make a point”
- How to create a line segment
- Maybe make the tour start at a certain point of the page that can auto open.
- There is to be a listener inside the UITour component in which it will listen for a ‘next’ click.
 - Vuetify and VueJS is what is to be used in the project
 - Look for VTooltip documentation
 - It is using the activator property.

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Reminders

Make sure to go over Vue and Firebase and looking up the documentation for **VToolTip**

Outlook/Main Goals

Our main goal with the tours is to essentially make a 'plan/blueprint' so that the tours can be created in the future.

We are making a **tool/workflow** that can easily create tours on the backend.

That is, we want to make sure it is easily codeable.

Would strongly like to have a record tours feature.

Not necessarily just creating a bunch of tours. MAKE A BLUEPRINT FOR CREATING TOURS.

We want to focus on making pop up boxes look pretty and how to move onto the next.

Possible branching to go into other tours within them could be something to think about.

Our very first main thing to deliver is to see how we can make a tour for the very simplest action: how to create a point.

These tours should help guide/teach students about the importance of things one may not think about. Such as the other angle that is created on the sphere, which is the outside angle.

If a user is signed in, we could possibly make a system that can keep track of tours that have been completed by the user.

Tests were able to check if an item was made but wasn't entirely sure where it was.

Make sure we have it hard coded first, START SMALL!!!!

But, in the end, they would like to make it interactive with the user so that it can take in user input to not need a next button.

Objects Panel

The tokens next to the items can be used to calculate many different expressions within the objects section of the left hotbar. These measurements get updated in real time when moved on the easel.

Main ideas for next

Build off of Dr. Dulimarta's presentation. Instead of a timer to move to the next, we make some kind of manually pressed button. How can we direct users to guide where they need to go?

"What do I need to put everywhere on every UI element to know where I can get to?"

Start thinking about where we should be going.

Next week we should be able to brainstorm and have our own instances so that we can start playing around with the code so we can mess around with it.

Look into the **VToolTip**, and get familiar with the documentation and the API of Vue itself.

We should be brainstorming about what we should be doing.

Office Hours for Dr. D @ C2415:

Tuesdays, Thursdays: 2-3pm

Wednesdays: 3-4pm

Office hours meeting with Dr. Dulimarta on getting the local build working 9/17:

Local Development Setup

Initial Setup

- Clone the repository directly into the project folder of your choice.
- This can be done through an IDE or through the terminal.
- When using the terminal, make sure it is opened in the folder where the project will be stored.
- Follow the setup instructions listed on the GitHub repository page.
- Bun is the recommended method for running the program because it is faster.
- Follow the installation instructions on the Bun website.
- Run the following command in the project folder:

```
bun install
```

Environment Configuration

- The Slack page contains the keys required to access the application.
- Create a file named `.env.local` and place the required keys inside it.
- Make sure the file is named `.env.local` and not `.env`. Otherwise, the keys may become public.
- The Firebase configuration file must also be created to access the site.
- The required Firebase code is provided through Slack.

Running the Application

- Run the following command from the project folder:

```
bun app:serve
```

- This will start the application locally.

Useful Git Commands

- git log — Displays the history of Git edits and commits.
- git checkout — Switches to another branch or revision.
- del node_modules — Deletes the existing node_modules folder when previous modules need to be removed or reinstalled.
- PowerShell can use Tab to autocomplete commands and file paths.

VS Code

- Install the Vue (Official) extension in VS Code for Vue development.

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Meeting Notes

Project Sprint Schedule:

Sprint 1

Week 1: Development environment setup, codebase exploration, GitHub/Linear workflow

Week 2: Tour prototypes and comparison of implementation approaches

Sprint 2

Week 1: Tour structure, data model, and framework design

Week 2: Basic reusable tour framework and first working tour

Sprint 3

Week 1: User action detection, completion conditions, and dynamic progression

Week 2: Reusable dynamic tour system and additional tours

Sprint 4

Week 1: User testing, UX refinement, and bug identification

Week 2: Core feature completion, testing, documentation, and stable demo build

Sprint 5

Week 1: Final testing, documentation, architecture diagrams, and final report draft

Week 2: Final report, demo video, poster, presentation, and final client review

Topics:

Code Exploration

- On Firebase, the minimum requirement will be to check whether a user has completed a tour.
- Poll requests will be addressed under Sprint 5.
- Team members will notify each other when they see merge requests and ensure that they are properly created and reviewed.
- We discussed the idea of adding a button directly to the tooltip to move through the next steps, rather than placing the button on the top side of the canvas.
- Markdown will be used for documentation, although Dr. Dulimarta may prefer to keep a README as well.
- The user manual is located under the docs directory.

AI Usage

- The team should not upload the entire cloned repository to Claude.- Dr. Dickinson described the use of AI as somewhat “fuzzy,” particularly because

AI-generated code may not always be maintainable.

- The role and acceptable use of AI are still somewhat in flux, with no strict boundaries established beyond not uploading the entire cloned repository.

Codebase Exploration

- localhost:8080/?features=beta can be used to access the site's beta transformation tools.
- Line 32 in src/main.ts contains information related to the beta tools.
- Vue components are divided into three main sections: HTML, TypeScript, and CSS.
- localhost:8080/tutorial opens the tour mode.
- The team is continuing to explore the existing anchors and how they function.

Goal for Next Meeting

- Each team member will create a prototype of one of the first tours, such as making a point or making a line segment.

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Meeting Notes

1. IP Waiver

- All group members will have their IP waivers signed in person at the meeting.
- The signed waivers will be scanned and uploaded digitally.

2. Tour Navigation, Instructions, UI Layout & Controls

- Dr. D discussed how a graphic designer was brought in during a summer program to redesign the site's layout and improve how users interact with it.
- It was emphasized that maintaining and honoring the site's existing look and design is important.
- We discussed file organization because the current demo tour is already becoming crowded. UITour.vue will likely need to be separated into multiple files to keep the code organized.
- We still need to determine how user input and actions will be detected. Dr. D should be able to help us work through this when he returns.

Main Overview

- The sponsor wants the Help tab to be opened and closed using a single icon, similar to the question-mark button in Desmos.
- When a user selects a tour, the Help tab will close and UITour/VTooltip will take over to guide the user to the appropriate UI elements.
- Once the user completes a tour, the completed tour will be logged.